

SUMMER INTERNSHIP TRAINING REPORT ON

**Intelligent Forensic Survey of Vehicular Motion and
Impact Zones Using Aerial Imagery**



SUBMITTED BY

Om Prakash Barik, Samrat Yadav, Anshuman Priyadarshi, Baibhav Kumar Mishra

FROM

Odisha University of Technology and Research, Bhubaneswar

UNDER THE GUIDANCE OF **SHRI AVEEK DUTTA , SC-'E'**,

**PROOF AND EXPERIMENTAL
ESTABLISHMENT, DRDO,
CHANDIPUR, ODISHA**

ACKNOWLEDGEMENT

We would like to extend our heartfelt gratitude to Mr. Aweek Dutta, Scientist E, Joint Director, PXE, DRDO, for his invaluable guidance and support throughout the Internship Training. His profound knowledge and expertise in the field of radar technology and his unwavering commitment to excellence have been instrumental in shaping this work. Mr. Dutta's insightful feedback and constructive suggestions have significantly enhanced the quality and depth of this report. His encouragement and motivation have been a constant source of inspiration, driving us to delve deeper into the subject matter and strive for accuracy and clarity in our project.

We are thankful for getting the opportunity to carry out Summer Internship Training at PXE, DRDO which had many overwhelming experiences from seeing the working of a Radar to witnessing the firing of ATAGS Gun, and MBT Arjun. The entire duration of our training was highly inspiring and knowledgeable and offered us a platform for enhancing our skills and using them to solve practical problems.

AN INTRODUCTION TO DRDO PXE

The Proof and Experimental Establishment (PXE) is a premier laboratory under the Defence Research and Development Organisation (DRDO) of India. Established in 1895 and located in Chandipur, Odisha, PXE is dedicated to the testing and evaluation of armaments and their components. Its primary mission is to ensure the reliability, accuracy, and safety of defense equipment through rigorous experimental procedures and proof tests.

PXE is equipped with state-of-the-art facilities and infrastructure to conduct a wide range of tests, including ballistic evaluation, range testing, environmental tests, and metallurgical analysis. The establishment plays a critical role in supporting the Indian Armed Forces by validating the performance and efficacy of various munitions, including artillery shells, rockets, and bombs.

Through continuous innovation and development, PXE contributes significantly to India's defense capabilities, ensuring that the armaments used by the military are both effective and safe. The establishment's work is pivotal in advancing India's strategic defense technologies, bolstering national security, and maintaining a robust defense posture.

CONTENTS

Lab Visit

1. Survey Section
2. Radar and Sensor Wing
3. Communication Wing
4. Weapon Wing
5. Imaging wing
6. Ammunition and Preparation wing

Practical Project

1. Objectives
2. Module 1: Vehicle Speed Detection using opencv
3. Module 2: Damage Assessment from Aerial Imagery
4. Module 3: Detection of Vehicle Anomaly through CCTV Footage
5. Conclusion
6. References

LAB VISIT

SURVEY SECTION

RT MAPS

In surveying, RT MAPS (Real-Time Maps) refer to the real-time generation of spatial maps and coordinates using modern surveying instruments—especially GNSS/GPS, Total Stations, and RTK systems (Real-Time Kinematic).

What is RT MAPS in Surveying?

RT Maps in surveying are digitally generated maps created in real-time during field surveys using:

- Satellite positioning systems (GNSS/GPS)
- RTK-enabled instruments for centimeter-level accuracy
- Total Stations
- Drones/UAVs with mapping sensors

These systems instantly convert raw sensor data into 3D maps, contours, and topographic models while you're still on site.

Key Technologies Behind RT Maps in Surveying

Technology	Role in RT Mapping
RTK GNSS	Provides real-time centimeter-level positioning accuracy
Total Station	Measures angles/distances; coordinates calculated instantly
GIS Software	Converts real-time measurements into usable maps and layers
Drones/UAVs	Capture aerial images used for 3D mapping and orthoimages
Data Collectors	Field tablets log and process data instantly

Process of Real-Time Mapping (RT Maps)

1. Set up GNSS/RTK or Total Station
2. Start real-time data collection with rover or prism

3. Live positioning and measurements are recorded
4. Mapping software (e.g., Trimble, Leica, Topcon) creates maps and terrain models in real time

Applications in Surveying

- Topographic surveys
- Land boundary mapping
- Construction layout
- Road/highway alignment
- Pipeline/cable mapping
- Agricultural land planning

Advantages of RT MAPS in Surveying

- Faster data collection
- On-site error checking
- Higher accuracy with RTK corrections
- Integration with CAD/GIS platforms
- Immediate availability of maps and reports

TOTAL STATION

A Total Station is a modern surveying instrument that integrates three key tools into one device:

1. Theodolite – Measures horizontal and vertical angles.
2. Electronic Distance Measurement (EDM) – Measures the distance from the instrument to a specific point.
3. Microprocessor with Data Recorder – Stores and processes measured data.

Working Principle

A total station emits an infrared or laser beam toward a prism reflector (or directly to the surface in reflectorless mode), then calculates:

- Distance using time-of-flight or phase shift
- Angles using precise encoders

These measurements are combined to compute coordinates (X, Y, Z) of the point relative to the instrument.

Applications

- Land surveying and topographic mapping
- Construction site layout
- Road and highway design
- Mining and excavation
- Archaeology
- Utility mapping (pipes, cables)

Advantages

- High accuracy (up to millimeter level)
- Quick data collection and automation
- 3D coordinate measurement
- Digital data storage and easy CAD integration
- Some models support GPS and wireless control

Components

- Telescope

- EDM unit
 - Control panel and display
 - Processor and internal memory
 - Battery
 - Optical/laser plummet
 - Tripod
-

Types

- Manual Total Station
 - Motorized/Robotic Total Station (can be operated remotely)
 - Reflectorless Total Station (no prism needed)
-

GPS AND DGPS

1. GPS (Global Positioning System)

GPS is a satellite-based navigation system developed by the U.S. Department of Defense. It enables users to determine their location, speed, and time using signals from at least 4 satellites.

How It Works:

- GPS satellites orbit Earth and continuously transmit signals with the time and location of the satellite.
- A GPS receiver on the ground calculates its distance from multiple satellites based on how long the signal took to arrive.
- Using trilateration, the receiver determines its exact position (latitude, longitude, altitude).

Accuracy:

- Typically around 5–10 meters in open sky.

2. DGPS (Differential GPS)

DGPS enhances GPS accuracy by correcting signal errors using a network of fixed ground-based reference stations.

How It Works:

1. A base station with a known, fixed location calculates errors by comparing the actual satellite signal with what it should be.
2. These correction signals are sent to nearby DGPS receivers (rovers) via radio or the internet.
3. The rover uses the corrections to improve its own GPS calculation.

Accuracy:

- Can improve GPS accuracy to 1–3 meters, sometimes even sub-meter level.

GPS vs DGPS: Comparison

Feature	GPS	DGPS
Accuracy	5–10 meters	0.5–3 meters (typical)
Reference Station	Not required	Required (base station setup)
Cost	Lower	Higher due to extra equipment
Use Cases	General navigation, mobile phones	Surveying, agriculture, mapping
Signal Correction	No	Yes (real-time or post-processed)

Applications

GPS:

- Smartphones and vehicle navigation
- Marine and aviation navigation
- Geotagging photos

DGPS:

- Land surveying
 - Precision agriculture
 - Marine navigation (harbors)
 - Autonomous vehicles and robotics
-

RADAR AND SENSOR WING

RADARS

What is RADAR?

RADAR (Radio Detection and Ranging) is a technology that uses radio waves to detect, locate, track, and identify objects at a distance.

Working Principle

1. A transmitter sends out radio waves (pulses).
 2. These waves bounce off objects (called targets).
 3. A receiver detects the reflected waves (echoes).
 4. The system calculates:
 - Distance (based on time delay of the echo)
 - Speed (using Doppler shift)
 - Direction and sometimes shape
-

Main Components

Component Function

Transmitter Generates high-frequency pulses

Antenna Sends and receives radio signals

Receiver Detects and amplifies return signals

Processor Interprets data to find range/speed

Key Parameters

- $\text{Range} = (\text{Speed of light} \times \text{Time delay}) / 2$
 - Resolution = Ability to distinguish between two objects
 - Doppler Effect = Used to calculate object's relative speed
-

Types of RADAR

1. Pulse RADAR – Sends short bursts of signals; used in air traffic and weather
2. CW RADAR (Continuous Wave) – Constant wave for speed measurement

3. Doppler RADAR – Detects motion and speed (used in police radars, meteorology)
 4. Phased Array RADAR – Electronically steered beam (used in defense)
 5. Synthetic Aperture RADAR (SAR) – High-resolution imaging from satellites/drones
-

Applications

- Military – Target tracking, missile guidance
 - Aviation – Air traffic control, weather detection
 - Marine – Ship navigation and collision avoidance
 - Meteorology – Rainfall and storm tracking
 - Automotive – Adaptive cruise control, collision detection
-

Advantages

- Works in all weather
 - Detects moving and stationary objects
 - Long range and high speed detection
-

1. Radar Range Equation

The Radar Range Equation calculates the maximum distance at which a radar can detect a target.

Basic Form (Monostatic Radar):

$$P_e = \frac{P_s G^2 \sigma \lambda^2}{(4\pi)^3 R^4}$$

Where:

- Rmax: Maximum radar detection range
- Pt: Transmitted power
- G: Antenna gain
- λ : Wavelength of the radar signal
- σ : Radar Cross Section (RCS) of the target
- Pmin: Minimum detectable signal power by receiver

Interpretation:

- Range increases with higher transmit power, antenna gain, and target RCS
- Range decreases with smaller RCS or low receiver sensitivity

2. Radar Cross Section (RCS)

Radar Cross Section (σ) is a measure of how detectable an object is by radar — it quantifies how much radar energy an object reflects back to the radar.

RCS is defined as:

$$\sigma = \lim_{r \rightarrow \infty} 4\pi r^2 \frac{|E_s|^2}{|E_i|^2}$$

Where:

- E_s : Scattered electric field
- E_i : Incident electric field
- r : Distance from the radar

Factors Affecting RCS:

- Size of the object
- Shape (sharp edges and flat surfaces reflect more)
- Material (metallic surfaces reflect better than plastic)
- Angle of incidence (direct hits reflect more strongly)

Typical RCS Values:

Object	RCS (σ) approx.
Commercial aircraft	10–100 m ²
Fighter jet (stealth)	0.01–1 m ²
Car	10–20 m ²
Bird	0.01 m ²
Human	1 m ²
Stealth drone	0.001 m ²

Relationship Between RCS and Radar Range:

- A larger RCS allows the radar to detect the object from farther away

- A smaller RCS (stealth technology) reduces detection range
-

Conclusion:

- The Radar Range Equation helps determine detection capability
 - RCS is a key factor in radar visibility — especially in military stealth design
-

PPTR RADAR

What is Pulse Phased Array tracking Radar?

A Pulse Phased Array tracking Radar (PPTR) is a radar system that:

- Transmits pulsed signals (not continuous)
 - Uses a phased array antenna, which steers the radar beam electronically (not mechanically)
 - Combines the advantages of pulse radar (range measurement) and phased array (rapid scanning and tracking)
-

How It Works:

1. Pulse Transmission: Radar emits short-duration, high-power radio wave pulses.
 2. Electronic Beam Steering: Instead of physically rotating the antenna, it changes the phase of the signal across many small antenna elements.
 3. Echo Reception: It receives reflected signals (echoes) to calculate range, speed, and direction of targets.
-

Key Components:

- Phased Array Antenna: Grid of individual radiating elements with phase shifters.
 - Pulse Modulator: Creates short-duration, high-power RF pulses.
 - Transmitter/Receiver Modules (TRMs): Located behind each antenna element.
 - Signal Processor: Handles beamforming, Doppler processing, and target tracking.
-

Advantages:

- Fast Scanning – Beam can jump between directions in microseconds
- Multiple Target Tracking – Can track dozens or even hundreds of targets at once

- High Reliability – No moving parts in beam steering
- Low Side Lobes – Better target discrimination and jamming resistance
- Long Range – Due to pulse energy and high gain

Applications:

- Military – Air defense (e.g., Patriot missile radar), target tracking, missile guidance
 - Naval radar systems
 - Space surveillance
 - Weather monitoring (advanced Doppler phased radar)
-

PPAR vs Traditional Radar

Feature	Traditional Radar	Pulse Phased Array Radar
Beam Steering	Mechanical	Electronic
Scan Speed	Slow (rotating)	Extremely fast
Reliability	Mechanical wear	Solid-state, high
Multi-target tracking	Limited	Advanced simultaneous tracking

DOPPLER RADAR

What is Doppler Radar?

A **Doppler radar** is a radar system that **measures the velocity of a target** by using the **Doppler Effect** — a change in frequency of the returned signal caused by the **motion of the object** relative to the radar.

Doppler Effect Principle

When a target moves:

- **Toward the radar** → Reflected frequency increases (compressed waves)
- **Away from the radar** → Reflected frequency decreases (stretched waves)

This **frequency shift** helps calculate the **speed of the object**.

How Doppler Radar Works

1. Transmits continuous or pulsed radio waves.
2. Waves hit a moving object and reflect back.

3. The radar receiver detects frequency shifts.
4. The system computes the **radial velocity** (speed toward/away from radar).

Types of Doppler Radar

Type	Description
CW Doppler Radar	Continuous Wave – only measures speed, no range info
Pulse-Doppler Radar	Sends pulses – measures both speed and distance
Frequency Modulated Continuous Wave (FMCW)	Combines CW and range detection; common in automotive radars

Key Features

- Can **detect moving targets in presence of stationary objects** (MTI – Moving Target Indication)
 - Filters out ground clutter or background
 - Provides **velocity information**, not just position
-

Applications

- **Weather forecasting:** Measures wind speed, storm movement
 - **Police speed guns:** Measure vehicle speed
 - **Aviation:** Track aircraft velocity, avoid collisions
 - **Military:** Target tracking, missile guidance
 - **Automotive:** Adaptive cruise control, collision avoidance
-

Velocity Calculation (Doppler Shift)

$$f_d = \frac{2v}{\lambda}$$

Where:

- f_d : Doppler frequency shift
- v : Relative velocity

- λ : Wavelength of the radar signal
-

Advantages

- Accurate velocity measurement
 - Works well in poor visibility
 - Can distinguish moving targets from stationary clutter
-

Limitations

- Cannot detect **stationary objects**
- Complex signal processing
- Sometimes limited in angle or cross-range velocity detection

SENSORS

What is a Sensor?

A **sensor** is a **device that detects or measures a physical quantity** (like temperature, pressure, motion, etc.) and **converts it into an electrical signal** for monitoring, control, or analysis.

Basic Working Principle

1. Senses a change in a physical parameter
 2. Converts it into an analog/digital signal
 3. Signal is processed/displayed or sent to a controller/system
-

Classification of Sensors

◆ Based on Measured Quantity:

Sensor Type	Measures
Temperature sensor	Heat/temperature
Pressure sensor	Pressure (force/area)

Sensor Type	Measures
Proximity sensor	Presence of objects
Light sensor	Light intensity
Motion sensor	Movement or acceleration
Humidity sensor	Moisture level
Gas sensor	Gas concentration
Force sensor	Applied force/load

◆ **Based on Output Signal:**

- **Analog sensors:** Continuous signal (e.g., thermistor, strain gauge)
- **Digital sensors:** Discrete output (e.g., IR sensor, encoder)

◆ **Based on Power Use:**

- **Active sensors:** Need external power to operate (e.g., ultrasonic sensor)
- **Passive sensors:** Operate without external power (e.g., thermocouple)

Common Sensor Technologies

- **Thermocouple / RTD** – temperature
 - **Strain gauge** – pressure/force
 - **Photodiode / LDR** – light
 - **Accelerometer / Gyroscope** – motion/orientation
 - **Piezoelectric** – vibration, force
 - **Hall Effect** – magnetic field
-

Applications of Sensors

Field	Use Case
Automotive	Airbags, speed sensors, parking assist
Healthcare	Heart rate, pulse oximeters, glucose monitors
Industrial	Machine automation, pressure/temp control
Consumer Electronics	Touchscreens, gyroscopes, facial recognition

Field	Use Case
Agriculture	Soil moisture, humidity, pH sensors
Robotics	Obstacle detection, line following

Advantages

- Enable automation
- Improve safety and precision
- Real-time monitoring

Limitations

- Can be affected by noise or environmental conditions
- May require calibration
- Cost and complexity increase with precision

IEPE TECHNOLOGY

What is IEPE?

IEPE stands for **Integrated Electronics Piezo-Electric**, a technology used in **dynamic sensors** such as:

- **Accelerometers**
- **Pressure sensors**
- **Force sensors**
- **Microphones**

It integrates the **signal-conditioning electronics (like a charge amplifier)** directly into the sensor body, simplifying connection and reducing noise.

Key Features of IEPE Sensors

- **Built-in amplifier** inside the sensor
 - Requires **constant current excitation** (typically 2–20 mA)
 - Uses a **single coaxial cable** for **power and signal**
 - Outputs a **low-impedance voltage signal**
-

How IEPE Works

1. A piezoelectric element inside the sensor generates a charge when pressure or force is applied.
2. The **internal electronics** convert this high-impedance charge into a **low-impedance voltage signal**.
3. The signal is sent through a **single cable** that also supplies the power (via constant current).

Excitation Requirements

- Voltage: typically **18–30V DC**
- Constant current: **2–20 mA**
(This is supplied by an IEPE-compatible signal conditioner or DAQ system.)

Advantages of IEPE Sensors

Feature	Benefit
Low output impedance	Long cable runs without signal loss
Integrated electronics	No external amplifier needed
Simplified wiring	Single cable for power & signal
Good signal-to-noise	Excellent for dynamic measurements

Limitations

- Cannot measure **static signals** (only dynamic events like vibration, shock)
- Requires **IEPE-compatible signal conditioners**
- Operates within a limited temperature range (usually up to 120–125°C)

Applications

- **Vibration monitoring**
 - **Shock and impact testing**
 - **Structural health monitoring**
 - **Modal analysis in mechanical engineering**
 - **Aerospace and automotive testing**
-

IEPE vs Charge Mode Sensors

Feature	IEPE Sensor	Charge Mode Sensor
Signal Conditioning	Built-in	Requires external amplifier
Cable Type	Coaxial, simple	Low-noise specialty cable
Impedance	Low	High
Cost	Generally lower	Higher due to external gear
Sensitivity to noise	Less	More

PRESSURE SENSORS

What is a Pressure Sensor?

A **pressure sensor** is a device that **detects pressure** (the force applied over an area) and **converts it into an electrical signal** that can be measured, displayed, or recorded.

Types of Pressure Sensors

Type	Measures	Common Use Cases
Absolute Pressure	Relative to vacuum	Barometric pressure, altimeters
Gauge Pressure	Relative to atmospheric pressure	Tire pressure, blood pressure
Differential Pressure	Difference between two points	Flow measurement, filter monitoring
Sealed Pressure	Relative to fixed reference	Industrial environments

Working Principle

Most pressure sensors use one of these methods:

- **Strain gauge** (resistive change when stretched)
- **Capacitive** (distance between plates changes)
- **Piezoelectric** (generates voltage under pressure)
- **Optical** (light beam deflection)
- **Piezoresistive** (semiconductor resistance change)

Common Technologies

Technology Description

Piezoresistive Silicon diaphragm changes resistance

Capacitive Diaphragm alters capacitance

Piezoelectric Crystals produce voltage under force

MEMS Micro-Electro-Mechanical Systems for miniaturized pressure sensors

 Applications

Field	Application
Automotive	Tire pressure, fuel injection, braking systems
Medical	Blood pressure monitors, ventilators
Industrial	Process control, fluid monitoring
Consumer	Washing machines, smartphones (altimeters)
Aerospace	Altitude sensing, cabin pressure control

Advantages

- Compact and versatile
- Real-time monitoring
- High sensitivity and accuracy

Limitations

- Requires calibration
- Sensitive to temperature, vibration
- May degrade over time in harsh environments

Output Types

- **Analog** (e.g., 0–5V, 4–20 mA)
- **Digital** (e.g., I2C, SPI, CAN)

IPG PRESSURE SENSOR

Here is your content **organized and formatted professionally** to paste directly into a technical or academic report:

B252 IPG Sensor by HPI GmbH (Austria)

Introduction

The **B252 Internal Pressure Gauge (IPG)** is a specialized pressure sensor developed by **HPI GmbH**, a leading Austrian manufacturer of high-precision ballistic and pressure measurement instruments. This sensor is designed to measure **gas pressure or blast effects** in **closed environments**, especially where traditional wired connections are not feasible.

Key Features

- Non-Cabled Operation:** Functions without direct cable connections, ideal for sealed chambers or hard-to-reach areas.
- Application-Specific:** Suited for use in **powder chambers**, **blast testing**, and **shock effect studies** on human bodies.
- Robust Design:** Engineered for high-risk environments such as **ballistic testing labs** and **defense research** centers.
- Flexible Integration:** Can be embedded or customized for specific research or industrial setups.

Typical Applications

- Ballistic and internal blast pressure measurements
- Development and **acceptance testing** in defense and aerospace fields
- Human body shock effect studies** in biomedical and safety research
- Routine quality testing** of explosive chambers or weapons systems

Related Components and Accessories

Component	Description
B252 IPG Sensor	Internal pressure gauge for gas or blast measurements in closed environments
B252 Piezo Gauge Kit	Includes the B252 sensor and BWF software for signal processing and analysis
B252 Tool Set	Dedicated tools for installation, calibration, and maintenance

Component	Description
Portable Data Processor	Mobile unit used to record and analyze pressure data from the IPG sensor

Relevance in Defense and Research

The **B252 IPG** has been cited in international defense tenders, such as one from the **Department of Defence Research and Development (India)**. This procurement included:

- The **B252-800 model**
- **Piezo gauge kit**
- **Tool set**
- **Portable data processor**

This highlights the sensor's importance in **military-grade testing** and **advanced scientific analysis**.

Conclusion

The **B252 IPG sensor** by HPI GmbH is a **highly specialized, reliable, and field-proven solution** for high-pressure measurement applications. Its ability to operate **without direct connectivity** makes it particularly valuable in **ballistic testing, closed-chamber experiments, and hazardous research environments**.

STRAIN GAUGE

Strain Gauge-Based Sensors: Bonded vs Unbonded

Strain gauge sensors are widely used for **measuring strain, force, pressure, torque, and weight**. These sensors work on the principle that a conductor or semiconductor's **electrical resistance changes when stretched or compressed**.

1. Bonded Strain Gauge Sensors

Definition:

A **bonded strain gauge** is a type where the strain-sensitive element is **glued (bonded)** directly onto the surface of the material (structure) under test using a **special adhesive**.

Construction:

- A **resistive foil or wire grid** is mounted on an insulating backing (e.g., paper or plastic film)
- Bonded using adhesives like epoxy or cyanoacrylate onto the surface of the structure

Working Principle:

- When the structure deforms due to applied force or pressure, the bonded strain gauge deforms with it
- This causes a **change in resistance**, which is measured using a **Wheatstone bridge**

Applications:

- Structural analysis of bridges, buildings, aircraft
- Load cells and pressure transducers
- Biomedical applications (e.g., prosthetic testing)

Advantages:

- Direct contact with the structure → higher accuracy
- Small and lightweight
- Suitable for static and dynamic strain measurement

Disadvantages:

- Installation requires surface preparation
 - Susceptible to environmental conditions (humidity, temperature)
 - Limited reusability
-

2. Unbonded Strain Gauge Sensors

Definition:

Unbonded strain gauges do not attach directly to the test structure. Instead, the strain is transmitted to the gauge through a mechanical linkage or structure.

Construction:

- Thin wires or foil are **suspended freely** between two fixed points (not glued to the test body)
- When the body deforms, the displacement moves the attachment points and stretches the wires

Working Principle:

- Displacement causes tension in the suspended wires
- This leads to a measurable **change in resistance** that correlates with the applied load or strain

Applications:

- Used in **force transducers, torque sensors, and accelerometers**
- Often used in **dynamic testing and industrial systems**

Advantages:

- Less affected by temperature and surface conditions
- Can be reused or recalibrated easily
- Often more rugged and long-lasting in industrial settings

Disadvantages:

- Less sensitive than bonded types
- Bulkier and more complex mechanically
- Not suitable for small or compact applications

 Comparison Table

Feature	Bonded Strain Gauge	Unbonded Strain Gauge
Mounting	Directly glued to surface	Freely suspended (not bonded)
Accuracy	High	Moderate
Size	Compact	Larger
Reusability	Limited	High
Affected by Environment	More	Less
Applications	Load cells, bridges, R&D	Force/torque sensors, accelerometers
Complexity	Simple setup	More mechanical parts involved

TYPES OF STRAIN GAUGES

Here's a clear breakdown of the differences and purposes of **T-rosette**, **rosette**, and **linear strain gauges**:

1. Linear Strain Gauge

Definition

A **linear strain gauge** is a **single-element sensor** that measures **strain in one specific direction**. It is the simplest and most commonly used strain gauge.

Key Characteristics:

- Measures **unidirectional strain** (along the length of the gauge)
- Must be aligned **exactly** with the direction of expected strain
- Cannot detect strain at other angles

Use Cases:

- Tensile/compressive strain measurement in one direction
 - Structural testing where load direction is well known
 - Basic materials testing
-

2. Rosette Strain Gauge

Definition

A **rosette strain gauge** is a combination of **three closely spaced linear strain gauges**, oriented at different angles, used to measure **multi-directional strain** on a surface.

Types:

Type	Configuration	Angles Between Gauges
Rectangular Rosette	Gauges at 0°, 45°, 90°	45° apart
Delta Rosette	Gauges at 0°, 60°, 120°	60° apart
3-Element Rosette	Any arrangement of 3 gauges	Usually custom angles

Why It's Used:

- Measures strain in **multiple directions**
- Allows calculation of:
 - **Principal strains**
 - **Maximum shear strain**
 - **Principal stresses** (if material properties are known)

Use Cases:

- Stress analysis where direction of strain is **unknown**
 - **Complex loading conditions** (e.g., pressure vessels, components under torsion)
-

3. T-Rosette Strain Gauge

Definition

A **T-rosette** is a type of **biaxial strain gauge** where **two linear strain gauges** are oriented **perpendicularly** (90° apart) — typically at 0° and 90°.

Key Characteristics:

- Designed for conditions where **principal stress directions are known**
- Easier to align and install than full rosettes
- Less data to process, but limited to known stress directions

Use Cases:

- **Industrial monitoring** in structures with known load paths
- Applications with **biaxial stresses** (XY-plane) but **fixed directions**
- **Long-term embedded sensors** in bridges, buildings, pipelines

Comparison Table

Feature	Linear Gauge	T-Rosette Gauge	3-Element Rosette
Number of Gauges	1	2	3
Measures Strain In	1 direction	2 perpendicular directions	3 directions
Used When Principal Direction is	Known	Known	Unknown
Principal Stress Calculation	No	Partially (if directions known)	Yes
Applications	Simple uniaxial loads	Biaxial with known axis	Multiaxial with unknown stress paths

ACCELEROMETER

What is a 3-Axis Accelerometer?

A **3-axis accelerometer** measures **acceleration forces** along three perpendicular axes:

- **X-axis (forward-backward)**
- **Y-axis (left-right)**
- **Z-axis (up-down)**

This allows full characterization of motion in **3D space**, making it highly suitable for **dynamic and high-impact environments** like weapon systems.

Role in Weapon Testing

In weapon testing, a 3-axis accelerometer is crucial for capturing:

- **Recoil forces**
- **Impact shock**
- **Flight trajectory**
- **Structural vibrations**
- **High-G launch environments**

These devices are embedded into **projectiles, warheads, barrels, or platforms** to record real-time acceleration data during firing, flight, and impact.

Key Features of 3-Axis Accelerometers for Weapon Testing

Feature	Description
High-G Tolerance	Must survive and measure extreme G-forces (often >10,000g)
Shock Resistance	Withstand explosive or ballistic shock environments
Compact Size	Miniaturized to fit inside bullets, missiles, shells
High Sampling Rate	Accurate capture of fast transients during firing and impact
Data Logging or Telemetry	Some models store onboard data, others transmit wirelessly
Triaxial Sensing	Detect motion in all directions for full event analysis

Common Technologies Used

- **MEMS (Micro-Electro-Mechanical Systems)** accelerometers (compact, robust)
- **Piezoresistive** accelerometers (for high-frequency/high-G testing)
- **Capacitive** accelerometers (general motion sensing)
- **Piezoelectric** accelerometers (for high shock and vibration)

Integration in Testing Setups

- Mounted inside **artillery shells, missiles, warheads, or on test fixtures**
- Connected to:
 - **Data acquisition systems**
 - **High-speed loggers**
 - **Telemetry modules**
- Often used in conjunction with:
 - **Gyroscopes (IMU)**
 - **Pressure sensors**
 - **High-speed cameras**

Applications in Defense Testing

Use Case	Purpose
Recoil Analysis	Measure backward acceleration on weapon firing
Projectile Dynamics	Track flight motion in 3D
Shock Testing	Determine survivability of electronics during firing
Warhead Testing	Understand dynamic loads during detonation
Drop Testing of Ammunition	Assess structural integrity after accidental drops
G-Force Characterization	Ensure systems survive expected launch accelerations

Example Devices

- **Dytran 3200B series** (shock-hardened piezoelectric)
 - **PCB Piezotronics 3501HT series** (MEMS, high-G)
 - **Honeywell QA2000** (piezoresistive, tactical applications)
 - **ADXL377 (Analog Devices)** – compact MEMS, up to $\pm 200g$
-

Conclusion

A **3-axis accelerometer** in weapon testing provides **crucial dynamic data** under extreme conditions. It helps defense engineers:

- Validate **design performance**
- Ensure **structural durability**
- Improve **aerodynamic behavior**

SIGNAL CABLES

🔌 Signal Cables Used in Sensors and Measurement Systems

Signal cables are **critical components** in sensor-based measurement systems. They **transmit electrical signals** from sensors to data acquisition or control units. The right cable ensures **signal integrity**, **accuracy**, and **electromagnetic compatibility (EMC)**.

1. Types of Signal Cables

a) Shielded Twisted Pair (STP)

- **Structure:** Two conductors twisted together with a shielding layer (foil or braided)
- **Purpose:** Reduces **electromagnetic interference (EMI)**
- **Used In:** Strain gauges, thermocouples, low-level analog signals

b) Unshielded Twisted Pair (UTP)

- Similar to STP but **without shielding**
- Less protection from noise – used for **short-distance or digital signals** in low-noise environments

c) Coaxial Cable

- **Structure:** Central conductor, insulating layer, shield, outer jacket
- **Excellent noise immunity** due to coaxial geometry
- **Used In:** IEPE sensors, vibration transducers, RF signals

d) Triaxial Cable

- Similar to coax but with **an extra shield layer**
- Superior for **low-noise applications** (e.g., charge mode piezoelectric sensors)
- Used where **charge amplifiers** are used in measurement chains

e) Multicore Cables

- **Several insulated conductors** bundled together
- Used for systems with **multiple sensors or control lines**
- Often used in industrial automation and multi-sensor DAQ systems

f) Fiber Optic Cable

- Transmits **optical signals**, immune to EMI
- Ideal for **long distances** and **high-voltage isolation**
- Used in **smart sensors**, **explosive environments**, and **high-voltage labs**

2. Key Cable Properties to Consider

Property	Importance
Shielding	Prevents noise, essential for analog signals
Cable capacitance	Affects high-frequency signal accuracy
Flexibility	Needed for moving parts or robotic arms
Connector type	Must match sensor output (e.g., BNC, LEMO, M8/M12)
Temperature rating	Must withstand sensor operating conditions
Chemical/Mechanical resistance	Important in harsh environments

3. Examples of Sensor-to-DAQ Cable Types

Sensor Type	Typical Cable Used	Notes
Strain Gauge	Shielded Twisted Pair	Requires Wheatstone bridge input
Thermocouple	Thermocouple-matched cable	Must match type (e.g., Type K)
IEPE Accelerometer	Coaxial cable (e.g., BNC)	Carries power + signal
Charge Output Sensors	Triaxial cable	Very sensitive to noise
Digital Sensors (I2C, SPI)	UTP / Shielded multiconductor	For short distances

Best Practices in Signal Cabling

- **Keep analog signal cables short** to reduce noise pickup
- Use **twisted and shielded pairs** for differential signals (e.g., strain gauges)
- **Avoid running signal cables next to power cables**
- Use **differential amplifiers** with twisted pair cables for noise rejection
- **Proper grounding and shielding** is essential for low-noise environments

COMMUNICATION WING

ATOMIC CLOCK

What is an Atomic Clock?

An atomic clock is a highly precise timekeeping device that uses the natural vibration frequency of atoms (usually caesium or rubidium) as its reference. Unlike mechanical or quartz clocks, atomic clocks rely on quantum transitions of atoms, making them the most accurate timekeeping systems in the world.

Working Principle

Atomic clocks operate based on the principle of resonance frequency. Atoms (like cesium-133 or rubidium-87) emit or absorb electromagnetic radiation at a very specific frequency when they change energy states.

For example:

- Cesium-133 atoms resonate at exactly:

9,192,631,770 cycles/second (Hz)

- This frequency is used to define one second in the International System of Units (SI).
-

Components of an Atomic Clock

- Atom Source – Caesium or rubidium vapor is used.
 - Microwave Oscillator – Generates a frequency near the atomic transition frequency.
 - Cavity Resonator – Directs microwaves toward the atom beam.
 - Detector – Measures whether atoms changed state at the expected resonance frequency.
 - Feedback Loop – Adjusts the oscillator to lock onto the atom's natural frequency.
-

Types of Atomic Clocks

Type	Description	Accuracy
Cesium Beam	Most widely used; defines SI second	~1 second in 100 million years
Rubidium Clock	Compact and low-power; used in GPS satellites	Less accurate than cesium

Type	Description	Accuracy
Hydrogen Maser	Extremely stable; used in deep-space tracking	Very high short-term stability
Optical Clocks	Newest type; uses optical transitions (e.g., strontium)	Even higher accuracy (under research)

Applications

- Global Positioning Systems (GPS) – Precise satellite timing
- Telecommunications – Synchronization of data networks
- Astronomy & Space Science – Deep-space tracking, pulsar timing
- Financial Markets – High-frequency trading time stamps
- Scientific Research – Tests of relativity, quantum physics
- National Time Standards – Used by NIST, BIPM, and other labs to define universal time

Why Atomic Clocks Are So Accurate

- Atomic transitions are unaffected by temperature, pressure, or aging (unlike quartz or mechanical oscillators)
- Stability is derived from unchanging natural constants
- Can measure time with uncertainty less than a nanosecond per day

Examples

- NIST-F2: A cesium fountain atomic clock operated by the U.S. NIST, accurate to within 1 second in 300 million years
 - ESA's PHARAO: A cold atom clock used in space (ISS)
-

WEAPON WING

Gun Carrier Mark I (“Mark I Gun”) – First Self-Propelled Artillery

Overview

The **Gun Carrier Mark I** was a British innovation during WWI—adapted from the Mark I tank chassis to transport and, in the case of the howitzer, fire heavy artillery on the battlefield

Design & Specs

- **Based on:** Mark I tank chassis, with a lowered profile and twin rear cabins for driver and commander
 - **Dimensions:** ~9.75 m long (with tail), ~2.4 m tall; weight ~28 tons
 - **Crew:** 4 in the vehicle + 8-man gun crew ()
 - **Engine:** Daimler petrol, 105 hp
 - **Speed & Range:** Top speed ~6 km/h, range ~38 km
-

Armament Capabilities

- Designed to carry either:
 - **60-pounder field gun**, or
 - **6-inch howitzer** ()
 - Only the howitzer was **practically fired** from the vehicle platform; other guns needed dismounting ()
 - Equipped with a single machine gun for self-defense and ~60 lbs of ammunition for primary gun ()
-

Purpose and Use

- Intended to rapidly **transport artillery forward** under heavy fire following infantry assaults ()
 - Officially the **first modern self-propelled gun**, although its battlefield deployment favored logistics over firing combat missions ()
 - 48 units built between mid-1917 and early-1918 ()
-

Challenges & Operational Notes

- **Slow and unwieldy:** turning required tractors or horses; top speed limited to walking pace ()

- **Vulnerable during gun redeployment:** gun crews exposed and at risk during dismounting ()
- **Mostly repurposed** as an ammunition and supply transport vehicle after initial deployment ()

Legacy

- Pioneered the concept of **self-propelled artillery**
- Influenced later **WWII SPG designs**
- Demonstrated tactical mobility and battlefield value, despite mechanical limitations

Quick Summary Table

Feature	Details
Crew / Armor	4 vehicle + 8 gun crew; unsprung suspension
Engine	Daimler 105 hp petrol
Speed & Range	~6 km/h; ~38 km
Armament	60-pounder / 6-inch howitzer + MG
Production	48 units (mid-1917 to early-1918)
Role	Transport, limited firing capability

Bofors Gun – Signature Swedish Artillery

SE Origin & Development

- Designed in the **1930s by AB Bofors** (Sweden); widely acclaimed for its performance in anti-aircraft roles .
- Entered service in **1934**, becoming the standard **intermediate-calibre** AA gun between small rapid-fire weapons and heavy, long-range systems ().

Key Variants

1. 40 mm L/60

- **Most iconic version**, with over **60,000 units produced** ().
-

- Specs:
 - Rate of fire: **120–140 rpm**, muzzle velocity **~850 m/s** ().
 - Effective ceiling: **~7,200 m**, horizontal range **~3,800 m** ().
- Licensed and built globally (Canada, US, UK, India, Sweden, etc.) .

2. 40 mm L/70

- Post-WWII development for improved range, rate of fire, and western deployment (~300 rpm).
- Adopted by ~40 nations, including naval platforms and vehicles ().

3. Naval & Heavy Variants

- **57 mm L/70** naval autocannon (Mk 1–3), modern with programmable ammo and radar ().
- **155 mm FH 77** towed howitzer and **15.5 cm Bandkanon 1** self-propelled gun, showing Bofors' wider artillery expertise ().

Performance & Capabilities

- Praised in Belgian trials (1935) for **twice the speed and triple the target hits** compared to rivals
- In WWII, equipped Allied navies and ground forces; U.S. produced ~60,000 units via Chrysler .
- U.S. M1 variant was widely used, and later mounted on AC-130 gunships.

Global Use & Legacy

- Used by at least **19 countries** pre-WWII; post-war platforms include **Canada, Australia**, and more ().
- In India, the **155 mm FH 77 “Bofors” howitzer** played a crucial role during **Kargil War**, aiding in recapturing high-altitude positions .
- Continues in service—several navies still operate L/60 and L/70 guns after modernization .

Performance Summary

Model	Calibre	Rate of Fire	Range	Notes
40 mm L/60	40 mm	120–140 rpm	~3,800 m	WWII stalwart
40 mm L/70	40 mm	Up to 300 rpm		Post-war alt, NATO-standard amplifiers

Model	Calibre	Rate of Fire	Range	Notes
57 mm L/70 (Naval)	57 mm	~220 rpm	~17 km	Modern upgrade with programmable munitions
155 mm FH 77	155 mm	Sustained 3 rpm; burst higher	~21–27 km	Towable howitzer used by India

Key Takeaways

- **Revolutionary design** for its time — fast, accurate, mobile.
- **Widespread licensing** ensured global proliferation (100+ operators).
- **Adaptable platform**: from ground and naval AA to artillery and gunships.
- **Enduring legacy**: Modern upgrades and Indian ammunition indigenization reflect ongoing relevance

ATAGS – Advanced Towed Artillery Gun System (India)

IN Overview

- A **155 mm/52-calibre towed howitzer**, designed for high-precision, long-range artillery ().
- Developed by **DRDO/ARDE** in partnership with **Tata Advanced Systems, Bharat Forge (Kalyani)**, and others ().
- First prototypes completed between **2013–2017**, production began in **2019**, and inductees include Indian Army and Armenia ().

Key Specifications

Feature	Description
Weight	~12 t (initially ~18 t) ()
Barrel Length	~8.06 m (L/52)
Crew	6–8 personnel
Elevation	–3° to +75° (servo-driven)
Traverse	±25° (servo-driven)
Recoil System	Electro-/magneto-rheological

Feature	Description
Breech	Screw-type
Range	HE-BB: ~48 km; rocket-assisted: ~80 km (RAMJET)
Rate of Fire	Burst: 5/60 s → up to 6 in 30 s; Intense: 10 in 2.5 min; Sustained: 60/hour
Mobility	Towed at 60–80 km/h; self-propelled at ~18 km/h via 95 kW APU
Automation	Auto-alignment, loading crane, servo controls, thermal sights, and ballistic computer

Advanced Features

- **All-Electric Drives** replace traditional hydraulics for enhanced reliability and reduced maintenance .
- Supports **Multiple-Round Simultaneous Impact (MRSI)** for high-impact effect .
- Integrated with **C4I systems** (e.g., ACCCS “Shakti”), **thermal sights**, **navigation**, and **muzzle-velocity radar** .
- Designed for **future long-range guided munitions (LRGM)** .

Production & Deployment

- Approved for **307 units** under a ₹6,900–7,000 Crore deal, domestically built (60% by Bharat Forge, 40% by TASL) .
- Already deployed to multiple artillery regiments, especially along key strategic borders.
- **Export interest:** Armenia ordered at least 84 units, with six delivered

Strategic Significance

- Replaces outdated 105 mm/130 mm and Bofors-era guns, bridging a long gap in the Indian arsenal .
- Enhances **artillery lethality**, **range capabilities**, and **accurate night-firing**.
- Strengthens **Make-in-India** defense objectives and signals **indigenous high-end technology** prowess

In Summary

The ATAGS is a **state-of-the-art**, **self-sufficient**, **highly automated**, and **long-range** towed howitzer. It combines indigenous design, advanced electronics, and far-reaching firepower—

ushering Indian artillery into the modern era and serving as a strong contender in the global market.

TANKS

1. T-72 Ajeya (Combat Improved Ajeya - CIA)

Overview:

- The **T-72 Ajeya** is India's licensed version of the Soviet T-72M1 main battle tank.
- Manufactured by **Heavy Vehicles Factory (HVF), Avadi**, it has formed the backbone of India's armored forces since the 1980s.

Specifications:

- **Armament:** 125 mm smoothbore gun with autoloader.
- **Protection:** ERA (Explosive Reactive Armor) Mk-II and thermal imaging sights.
- **Mobility:** Powered by a 780 hp V-46 diesel engine.
- **Fire Control:** Night vision, laser rangefinder, and computerized fire control (in upgraded versions).

Upgrades in CI-Ajeya:

- **ERA panels, Fire Detection and Suppression System**, GPS-based navigation.
- **Thermal sights** and **communications enhancements** for modern battlefield performance.

Relevance:

- Despite age, still operational (~2,400 units).
 - Future Combat Vehicle (FRCV) program aims to replace it.
-

2. Arjun MBT (Main Battle Tank)

Background:

- Indigenous tank developed by **DRDO's Combat Vehicles Research and Development Establishment (CVRDE)**.
- Initiated in 1974 to reduce dependence on foreign tanks.
- Took over two decades to mature into a combat-ready platform.

Mk-1 & Mk-1A Variants:

Arjun Mk-1:

- Introduced in early 2000s, used in limited numbers (124 tanks inducted).
- Features:

- 120 mm rifled gun (with capability to fire **LAHAT ATGM**).
- **Kanchan composite armor** (developed to defeat APFSDS rounds).
- Fire control with thermal sights, stabilized gun, and night fighting capability.

Arjun Mk-1A:

- Successor with 72+ enhancements over Mk-1.
- Includes:
 - Remote-controlled weapon station.
 - Advanced **hunter-killer capability**.
 - Upgraded **fire control, communication systems, mine plough, and track-width mine plow**.

Mobility:

- Original engine: MTU MB 838 Ka-501 (German); future versions to use **indigenous 1500 hp DATRAN engine**.

Performance:

- In 2010 comparative trials, **Arjun outperformed T-90S** in many parameters (accuracy, mobility, firing).
- However, it is **heavier (68.6 tons)** than ideal for some Indian terrain, which delayed full acceptance.

3. Tank-Ex (NAKUL)

Concept:

- Experimental hybrid MBT combining:
 - **T-72 chassis** (Ajeya) – lightweight, mobile.
 - **Arjun's turret and 120 mm gun** – advanced firepower.
- Aimed to modernize existing T-72 platforms with Arjun's firepower.

Features:

- 120 mm rifled gun compatible with **LAHAT ATGM**.
- Hydropneumatic suspension for superior mobility.
- Protection: Composite and ERA; crew of 4.
- Range: ~500 km; Speed: ~60 km/h.

Outcome:

- Rejected after trials in desert conditions.
- Possibly due to weight issues and logistics mismatches with existing platforms.

Summary Table

Tank	Origin	Main Gun	Engine Power	Armor Type	Status
T-72 Ajeya	Soviet/India	125 mm (smooth)	780 hp	ERA Mk-II, Steel	Operational, being upgraded
Arjun Mk-1A	Indigenous	120 mm (rifled)	1500 hp (planned)	Kanchan + ERA	In limited production
Tank-Ex	Hybrid	120 mm (rifled)	1000+ hp	Composite + ERA	Rejected (prototype stage)

Strategic Importance

- **T-72 Ajeya:** Forms core of armored divisions but reaching obsolescence.
- **Arjun MBT:** Demonstrates India's capability in armored vehicle design; ideal for plains and semi-desert warfare.
- **Tank-Ex:** Valuable R&D experiment; lessons used in Arjun and other future systems.

IMAGING WING

High-speed cameras play a **critical role in weapon testing** due to their ability to capture events that occur in **microseconds to milliseconds**, which are far too fast for the human eye or conventional cameras to observe. Here's a detailed breakdown of their use and importance:

Applications in Weapon Testing

1. Ballistics Analysis

- **Projectile motion:** Tracking bullets, missiles, or shrapnel in flight.
- **Muzzle flash and recoil:** Analyzing combustion gases, muzzle blast, and weapon recoil dynamics.
- **Impact behavior:** Capturing how a projectile deforms, penetrates, or fragments upon hitting a target.

2. Explosive Testing

- **Detonation events:** Observing shockwave propagation, explosive fragmentation, and energy dispersal.
- **Explosive lens testing:** Crucial in nuclear weapons R&D (e.g., implosion-type devices).
- **Charge timing:** Measuring exact detonation sequences of multi-point explosive systems.

3. Armor and Material Testing

- **Penetration resistance:** Evaluating how materials respond to high-velocity impacts (e.g., ceramic, Kevlar, composite armor).

- **Failure modes:** Studying cracks, spalling, and deformation under extreme stress.

4. Rocket and Missile Testing

- **Launch dynamics:** Monitoring plume behavior, staging events, and guidance control.
- **Separation events:** Assessing booster or warhead separation with sub-millisecond precision.

5. Electronic Firing Systems

- Ensuring proper **timing synchronization** between electrical components and mechanical reactions.

Camera Features Needed

Feature	Importance
Frame rate (e.g., 10,000–1,000,000+ fps)	To capture fast events clearly
Resolution (balance with frame rate)	Important for detail (e.g., 1280x800 at 10k fps)
Shutter speed	To avoid motion blur (<1 μ s often needed)
Trigger system	Precise synchronization with test events
Ruggedness	Withstands harsh environments (shock, vibration, heat)
Data handling	High-speed data offloading and storage

Common High-Speed Camera Brands

- **Photron** – Widely used in defense and industrial testing.
 - **Vision Research (Phantom cameras)** – Known for extreme frame rates and rugged builds.
 - **iX Cameras** – Used in explosive and shockwave analysis.
 - **Shimadzu** – Known for ultra-high-speed up to millions of fps.
-

Example Use Case: Bullet Impact on Body Armor

1. High-speed camera is triggered just before bullet leaves barrel.
2. Captures entire trajectory to impact (e.g., 50,000 fps).
3. Post-analysis measures:
 - Bullet deformation
 - Armor delamination
 - Backface signature on ballistic gel

AMMUNITION PREPARATION WING

1. Detailed Classification of Propellants

A. Single-Base Propellants

- **Composition:** Primarily nitrocellulose (NC), also called guncotton.
 - **Characteristics:**
 - Moderate energy release.
 - Relatively cool flame (reduces barrel wear).
 - Stable under varying temperature conditions.
 - Used where performance requirements are moderate.
 - **Application:**
 - Historically used in early tank guns and field artillery.
 - Still in use for small arms, mortars, and training ammunition.
-

B. Double-Base Propellants

- **Composition:** Nitrocellulose (NC) + Nitroglycerin (NG).
- **Characteristics:**
 - Higher energy than single-base.
 - Hotter burning—can lead to increased barrel erosion.
 - More powerful, suitable for high-velocity applications.
 - More sensitive to handling and temperature.

- **Examples:**
 - JA-2 (used in the M829 series of 120mm NATO tank ammunition).
 - NQ/QC compositions used by various countries.
-

C. Triple-Base Propellants

- **Composition:** Nitrocellulose + Nitroglycerin + Nitroguanidine.
 - **Characteristics:**
 - Nitroguanidine lowers flame temperature significantly.
 - Offers high energy but reduces barrel wear.
 - Preferred in long-barrel systems and for longevity.
 - Produces less muzzle flash—tactical advantage at night.
 - **Applications:**
 - Long-range artillery like 155mm or 152mm howitzers.
 - Naval guns and some rocket-assisted munitions.
-

D. Composite/Low Vulnerability Propellants (LOVA)

- **Composition:** Binder (e.g., polyurethane) + oxidizer (e.g., ammonium perchlorate) + energetic plasticizer.
- **Characteristics:**
 - Mechanically strong and thermally stable.
 - Lower sensitivity to impact and heat—safer to handle.

- Customizable burn profiles.
 - **Use Cases:**
 - Artillery systems like the Indian ATAGS.
 - Next-gen tank shells and modular charge systems.
-

2. Propellant Forms in Artillery and Tank Systems

Form	Description	Typical Use
Tubular grains	Hollow cylinders; control surface area burning	Medium to large-caliber rounds
Flake or sheet cut	Thin slices, often irregular	Small arms and some legacy artillery
Ball powder	Round granules; consistent performance	Small-caliber rounds
Extruded rods	High mechanical strength	High-pressure tank rounds
Modular bags	Pre-formed cylindrical charges in bags or cases	155mm artillery (e.g., M777, ATAGS)

3. Propellant Systems by Weapon Type

A. Tank Guns (e.g., NATO 120mm smoothbore, Indian Arjun 120mm)

- Propellant is contained inside a combustible cartridge case.
- Modular or single-piece ammunition used.
- Uses double-base or LOVA-type propellants.

- Typical Example:
 - M829A3 APFSDS uses JA-2 propellant in a fully combustible cartridge.

B. Artillery Guns (e.g., 155mm howitzers like M777, K9 Vajra, Dhanush)

- Uses separate loading of projectile and charge system.
- Charge systems:
 - Bagged charges (e.g., M119, M203A1).
 - Modular Artillery Charge System (MACS) with ignition tubes.
- MACS Modules (like M231/M232A1):
 - Clean-burning, insensitive to temperature variation.
 - Reduced logistics and safer handling.

C. Rocket-Assisted Projectiles (RAP)

- Hybrid: uses conventional propellant + rocket motor.
 - Uses solid composite propellants (e.g., hydroxyl-terminated polybutadiene [HTPB]).
 - Extends range (e.g., 155mm ERFB-RAP shells).
-

4. Advancements & Research Trends

A. Insensitive Munitions (IM)

- Designed to resist accidental detonation from fire, impact, or shock.
- Use LOVA and other high-stability formulations.

B. Electrothermal-Chemical (ETC) Propulsion

- Uses plasma to ignite the propellant more efficiently.

- Enhances muzzle energy and velocity control.
- Under development for future tanks and artillery.

C. Environmentally Friendly Propellants

- Efforts underway to reduce heavy-metal residues and toxic byproducts.
- Green propellant research includes nitroguanidine alternatives and non-toxic plasticizers.

Example Systems

System	Propellant Type	Notes
M1 Abrams (USA)	JA-2 (Double-base)	Combustible case cartridge; high-velocity APFSDS
K9 Vajra (Korea/India)	Modular Bag Charges (Triple-base / LOVA)	Uses conventional bag charges with safe ignition tubes
ATAGS (India)	Modular Charge System	Advanced Indian artillery system using locally developed propellants
2S19 Msta (Russia)	Multi-zone Bag Charges	Russian howitzer using zone-adjusted powder loads

PRACTICAL PROJECT

OBJECTIVES

- VEHICLE SPEED DETECTION USING OPENCV
- VEHICLE ANAMOLY DETECTION
- STRUCTURAL DAMAGE DETECTION FROM AERIAL IMAGERY

MODULE 1

Vehicle speed Detection Using opencv

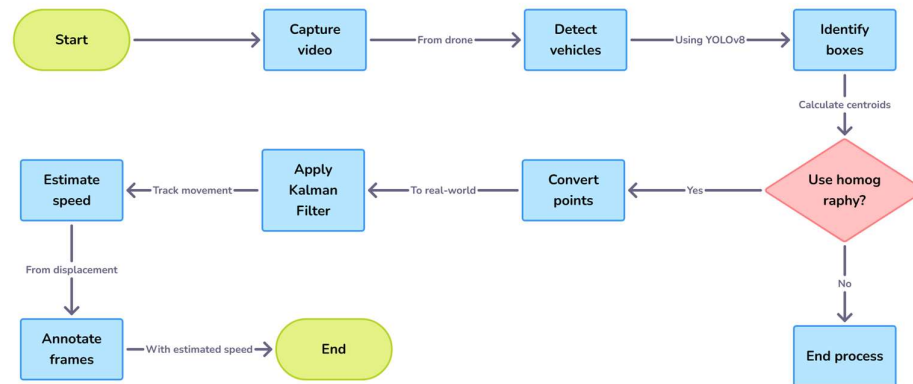
1.1 Introduction to Problem: The increasing number of vehicles on roads has led to rising traffic violations, including over-speeding, which in turn contributes significantly to accidents and road fatalities. Traditional speed detection systems such as radar guns and inductive loop sensors are either fixed in position or expensive to deploy over wide areas. The need for an intelligent, scalable, and aerially operable speed detection system has become critical. By leveraging drones and real-time video analysis, we aim to detect and estimate the speed of multiple vehicles simultaneously using computer vision and AI techniques.

1.2 Methodology: The process begins with capturing video footage from a drone-mounted camera. This footage is analyzed frame-by-frame using an object detection model (YOLOv8) to detect vehicles. Once a vehicle is detected, its position is tracked over subsequent frames. The speed is then estimated using the distance traveled by the object in the real-world plane divided by the time elapsed. To map pixel movement to actual ground distance, a perspective transformation is applied using homography. Further, to reduce noise and increase estimation accuracy, we apply a Kalman Filter to smooth out the predicted positions and velocities of vehicles.

Steps in the pipeline include:

- Capture video from drone
- Detect vehicles using YOLOv8
- Identify bounding boxes and calculate centroids
- Use homography to convert image points to real-world coordinates
- Apply Kalman Filter to track centroid movement over time
- Estimate speed from displacement and time difference
- Annotate the video frames with estimated speed

Flowchart



Made with Vally

Fig 1.1: Process flow for speed detection

1.3 Description of Algorithms:

- **YOLOv8 (You Only Look Once Version 8)**

YOLOv8 is the latest evolution in the YOLO (You Only Look Once) family of real-time object detection algorithms. Unlike traditional object detection methods that rely on a sliding window or region proposals, YOLO formulates the task as a single regression problem. It directly predicts bounding boxes and class probabilities from the input image in one forward pass through the network. This architecture results in extremely fast inference while maintaining high accuracy. YOLOv8 further enhances this framework with advanced features such as anchor-free detection, improved backbone networks (like CSPDarknet), better label assignment strategies, and more efficient post-processing.

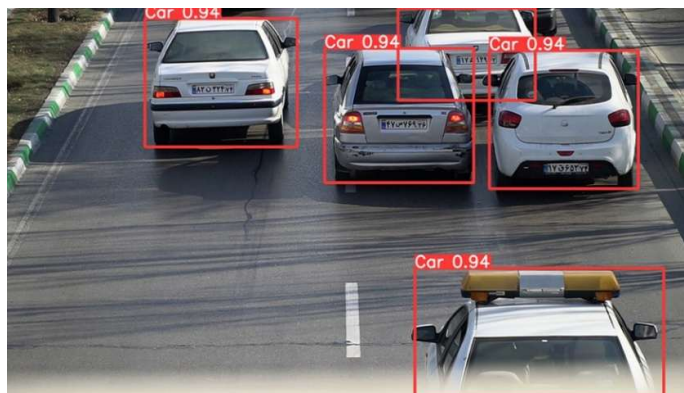


Fig 1.2: Object detection using YOLOv8

In this project, YOLOv8 is trained to detect various types of vehicles such as cars, buses, motorcycles, and trucks within aerial images or video frames captured by a drone. The model's high speed and efficiency enable it to perform in real time, processing each frame with minimal latency. This makes it suitable for continuous traffic surveillance and vehicle tracking in dynamic environments.

- **Homography Transformation**

Homography refers to a projective transformation that maps points from one plane in an image to another plane, typically using a 3×3 matrix. This transformation is especially useful when analyzing images captured at an angle, such as those taken by drones, where the camera's viewpoint creates perspective distortion.

In this project, homography is used to convert the oblique aerial view into a top-down (bird's-eye) view. This involves manually selecting four or more points on a planar surface (typically the road), and using these points to calculate the homography matrix. Once obtained, this matrix is used to map the image-space coordinates (like the centroid of a detected vehicle) to real-world coordinates on the road surface. This correction is crucial for accurate speed estimation, as it allows the calculation of true displacement over time in metric units rather than distorted image pixels.



Fig 1.3: Perspective transform using homography transformation

- **Kalman Filter**

The Kalman Filter is a recursive estimator widely used in tracking and navigation applications. It operates in two steps: prediction and correction. During the prediction step, the filter uses the object's previous state (position and velocity) to estimate the next state. In the correction step, it updates this estimate using new measurements and accounts for uncertainties (noise) in both the prediction and the measurements.

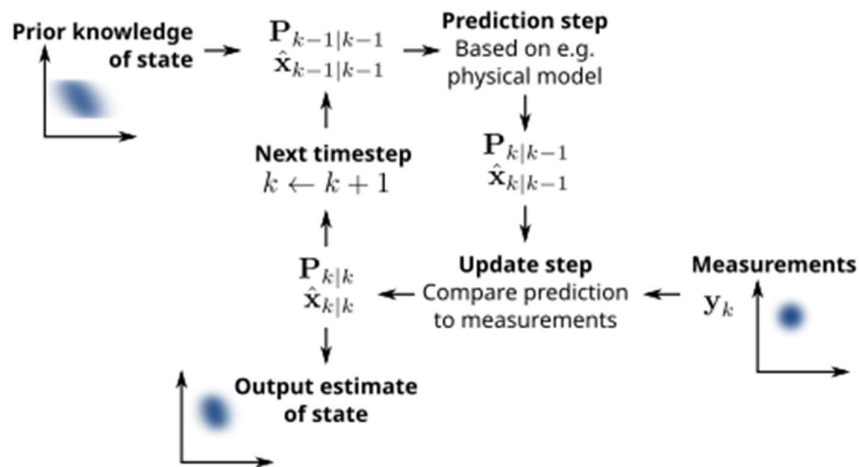


Fig 1.4 Kalman filter Algorithm

In this project, the Kalman Filter is applied to track the motion of each vehicle across frames. It predicts the position of a vehicle even when it is temporarily occluded or lost, such as when passing under a bridge or being blocked by another vehicle. This enables smoother and more consistent tracking. Additionally, since speed is computed from displacement over time, the filter helps in stabilizing the estimation, especially in noisy environments or when frame rates vary.

- **OpenCV (Open Source Computer Vision Library)**

OpenCV is an essential toolkit for image and video processing and is extensively used throughout this project. It supports a wide range of operations required for the full pipeline, including:

- **Reading Video Feeds:** OpenCV allows frame-by-frame extraction from live or recorded drone footage.
- **Visualization:** It provides functions to draw bounding boxes, labels, centroids, and trajectory lines on the frames to illustrate detection and tracking outputs.
- **Perspective Transformations:** OpenCV includes built-in functions to calculate and apply homography matrices for mapping image coordinates to ground-plane coordinates.
- **Preprocessing:** Functions such as grayscale conversion, Gaussian/median filtering, thresholding, and edge detection (e.g., Canny) help enhance important features while suppressing noise.

- **Feature Extraction:** OpenCV supports contour detection, keypoint identification, and geometric measurements, which are integral to damage analysis.
- OpenCV's flexibility and wide range of modules make it ideal for integrating all stages of detection, tracking, and visualization in a single, efficient workflow.
- **Speed Estimation Formula**
 Vehicle speed is calculated using the classical physics formula:

$$\text{Speed (km/h)} = (\text{Displacement in meters} / \text{Time interval in seconds}) \times 3.6$$
 - In the context of this project:
 - Displacement is derived by computing the difference in vehicle positions between frames after applying the homography transformation to convert pixel locations into real-world distances.
 - The time interval is based on the video frame rate, e.g., 1/30 seconds for a 30 FPS video.
 - The multiplication by 3.6 converts the result from meters per second (m/s) to kilometers per hour (km/h).

1.4 Results: We tested the system on various drone footages over highways. Vehicles were successfully detected and tracked using YOLO and Kalman Filters. The estimated speeds closely followed the ground-truth values with an average error margin of ± 2.0 km/h.

- Accuracy of vehicle detection: ~80%
- Average speed estimation error: ± 2 km/h
- Vehicles tracked concurrently: Up to 8
- Frame rate: 25 FPS real-time processing

Graphical results (e.g., Kalman-filtered speed curves) demonstrate the advantage of smoothing over raw speed estimates which are often noisy due to detection jitter.

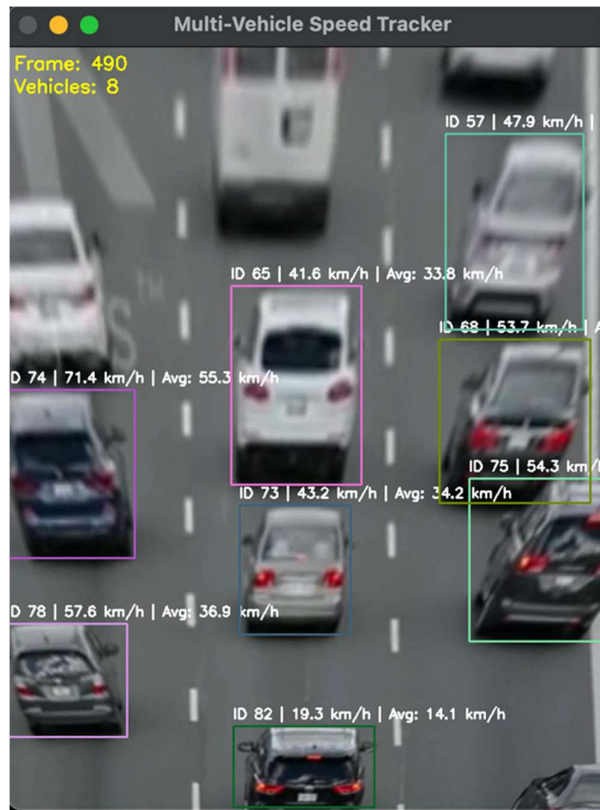


Figure 1.5 Speed detection results

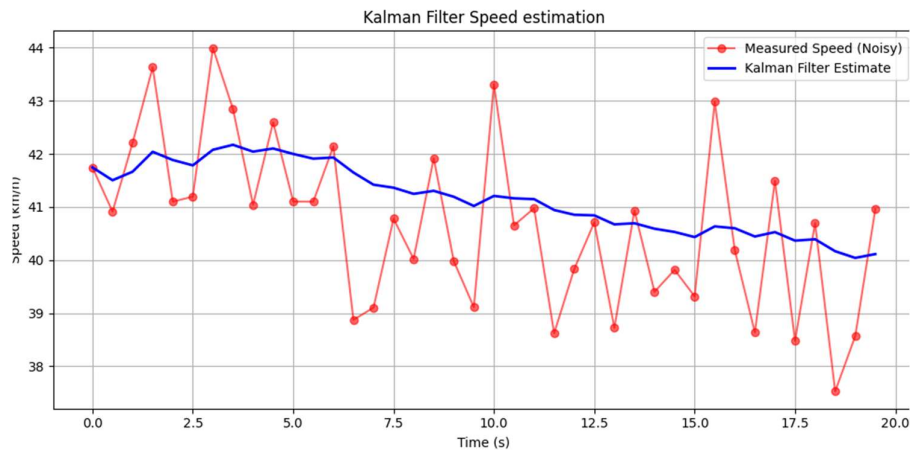


Figure 1.6 Kalman filter for noise reduction

1.5 Problems Faced:

- Real-world scaling required accurate selection of homography points, which is sensitive to user error
- Detection accuracy dropped under poor lighting or rainy weather

- Motion blur in video frames made centroid tracking difficult
- Plate detection occasionally misclassified due to low-resolution input
- Manual initialization of bounding boxes required in some scenarios

1.6 Future Scope:

- Fully automate homography estimation using deep learning or SLAM-based systems
- Integrate high-resolution cameras and thermal sensors for night-time operation
- Implement drone flight controller integration to allow real-time vehicle following based on number plate match
- Enable swarm drone architecture for city-wide traffic speed enforcement
- Extend the platform to predict driver behavior and potential collision risks using trajectory analysis

MODULE 2

Damage Assessment from Aerial Imagery

2.1 Introduction to Problem: Rapid and accurate assessment of structural damage after bombings, natural disasters, or urban conflict is essential for timely emergency response and strategic planning. Manual surveys are not only time-consuming and hazardous but also prone to inconsistencies. The increasing availability of drones offers an efficient platform to capture high-resolution imagery of affected regions. This project aims to develop an automated system that utilizes aerial images to detect and highlight damaged regions by comparing current visuals with satellite or pre-disaster imagery using advanced image processing techniques.

2.2 Methodology: This method involves processing before and after aerial images of the same region to identify and highlight structural damage caused by events such as natural disasters. The approach leverages a combination of image preprocessing, filtering, feature extraction, image matching, and statistical estimation techniques to robustly detect and estimate damage confidence. The key components of this pipeline are resizing, grayscale conversion, filtering, thresholding, contour detection, keypoint matching using SIFT and Brute-Force KNN, and confidence estimation through the Kalman filter.

Steps:

1. Input before and after aerial images.
2. Resize images, align them spatially, and convert them to grayscale.
3. Apply Gaussian and Median filters to remove noise and preserve edges.
4. Perform thresholding to convert the image to binary form.
5. Use contour detection to extract the boundaries of potential damaged areas.
6. Detect key points using SIFT in both images.
7. Match the key points using Brute-Force KNN and apply a ratio test to retain strong matches.
8. Identify regions with missing or displaced matches as damaged.
9. Use Kalman Filter covariance estimation to assign confidence to each detected damaged region.
10. Highlight the final damaged regions on the output image.

Flowchart

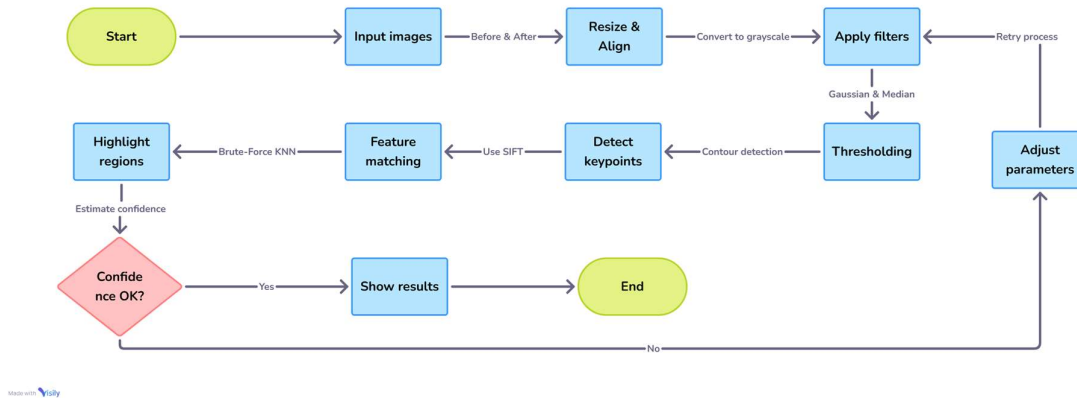


Fig 2.1: Process flow for damage detection

2.3 Description of Algorithms:

- **Gaussian and Median Filtering**

These filters are essential preprocessing steps to reduce noise while preserving important image details.

- **Gaussian Filter:**

This filter uses a Gaussian kernel to smooth the image, reducing high-frequency content such as random noise. It works by averaging the surrounding pixels weighted by the Gaussian distribution, which ensures that pixels closer to the center have more influence. This is particularly useful for removing subtle image artifacts while retaining overall structural shape. The Gaussian filter is defined as:

$$G(x, y) = (1 / 2\pi\sigma^2) * \exp(-(x^2 + y^2) / 2\sigma^2)$$

where σ controls the extent of the smoothing.



Figure 2.2 Gaussian filter

- Median Filter:

Unlike the Gaussian filter, which uses averaging, the Median filter replaces the central pixel value with the median value from its neighborhood. This approach is especially effective at removing salt-and-pepper noise while maintaining sharp edges. It does not blur boundaries, making it preferable for applications involving structural analysis.



Original Image



with Median Filter

Figure 2.3 Median filter

- **Thresholding and Contour Detection**

These are used to identify and localize potential damaged areas from difference images or filtered results.

- Thresholding:

Thresholding converts a grayscale image to a binary image using a fixed or adaptive threshold. All pixels above the threshold are assigned one value (e.g., white), and those below are assigned another (e.g., black). This step highlights significant intensity differences, which are often indicators of damage.

- Contour Detection:

Once the binary image is obtained, contour detection algorithms like the Suzuki-Abe algorithm trace the outlines of white regions. Contours are useful to localize and extract features such as area, shape, or bounding boxes for potential damage regions.

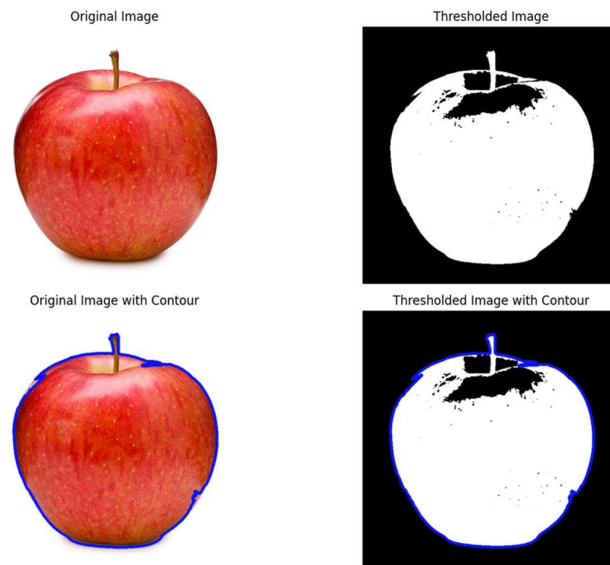


Figure 2.4 Thresholding and Contour detection

- **SIFT (Scale-Invariant Feature Transform)**

SIFT is a key algorithm for detecting and describing local features in images, with invariance to scale, rotation, and illumination changes.

- **Keypoint Detection:**
SIFT identifies keypoints by locating extrema in Difference-of-Gaussian (DoG) pyramids across multiple scales. Each keypoint represents a distinctive and repeatable pattern such as corners, blobs, or edges.
- **Descriptor Generation:**
For each keypoint, SIFT computes a 128-dimensional vector that captures the gradient orientation and magnitude in the surrounding region. These descriptors are highly distinctive, enabling robust matching across different images.
- **Advantages:**
SIFT is particularly useful in aerial imagery because it handles changes in perspective and scale—common challenges when comparing before-and-after images



Figure 2.5 SIFT algorithm for image matching

- **Brute-Force KNN Matching**

To compare features between the two images, feature matching is performed using a brute-force approach combined with the k-nearest neighbor (KNN) method.

- Brute-Force Matcher:

This algorithm compares each SIFT descriptor from the first image with all descriptors in the second image using Euclidean distance. The two closest matches ($k=2$) are retrieved for each descriptor.

- Lowe's Ratio Test:

To ensure robustness, a ratio test is applied where a match is accepted only if the distance to the best match is significantly less than to the second-best match. Typically, a threshold of 0.75 is used. This helps filter out ambiguous or incorrect matches.

- Application to Damage Detection:

Areas with fewer or no matches are likely to have undergone structural changes, such as collapsed buildings or newly exposed surfaces.

- **Kalman Filter for Confidence Estimation**

The Kalman Filter is a recursive estimator that is used here to quantify the reliability of the detected damage regions.

- Prediction and Update:

The Kalman filter operates in two phases: a prediction step where it forecasts the state (e.g., likelihood of damage in a region), and an update step where it

incorporates new measurements (e.g., change in pixel intensity, missing features).

- Covariance Estimation:

The Kalman filter also maintains a covariance matrix representing the uncertainty of the state. This allows the system to highlight damage zones with high confidence and suppress those with high uncertainty or noise.

- Benefits in Aerial Analysis:

By modeling temporal and spatial consistency, the Kalman filter helps stabilize detections, smooth out sporadic errors, and improve overall reliability of the final output.

2.4 Results: The damage detection model was evaluated on urban infrastructure datasets post simulated bombings and disaster scenes. The following outcomes were observed:

- Damage localization accuracy: ~80%
- Average false positives: < 7%
- Edge-preserving damage visualization overlay



Figure 2.6 Damage overlay

A visual overlay highlighted in red showed structural regions that have undergone visible destruction. These included buildings, roads, and boundary walls.

Sample results include:

- Matched keypoints between pre and post images (using SIFT)
- Highlighted damage overlays (red zones)

- Aligned aerial image and satellite reference



Figure 2.7 Zoomed in view of most damaged area

2.5 Problems Faced:

- Varying camera angles and lighting conditions caused poor feature matching
- Vegetation and seasonal changes introduced false differences
- Cloud shadows and brightness changes led to noise in subtraction
- Limited availability of pre-disaster satellite images for alignment

2.6 Future Scope:

- Integrate change detection with satellite APIs (e.g., Google Earth Engine)
- Use deep learning-based segmentation models (like UNet or Mask R-CNN)
- Develop real-time damage assessment system for military surveillance drones
- Add severity classification to damage regions (minor, moderate, severe)
- Create 3D reconstructions of damaged zones using photogrammetry

MODULE 3

Detection of Vehicle Anomaly through CCTV Footage

3.1. Introduction to the Problem

With the increasing use of CCTV in urban surveillance, manual monitoring of vehicular activities becomes infeasible and error-prone. Detecting suspicious vehicle behavior, such as repeated entries, loitering, or anomalies in entry/exit time, is critical for crime prevention and traffic management. This project automates the process by using computer vision and machine learning for anomaly detection in vehicle behavior based on number plate analysis.

3.2. Methodology

The system works through the following stages:

1. Vehicle detection using YOLOv8
2. Region of Interest (ROI) extraction for number plates
3. Preprocessing and edge detection using OpenCV
4. License plate OCR using Tesseract and EasyOCR
5. Logging of vehicle entries and timestamps
6. Anomaly detection based on frequency, time patterns, and OCR confidence
7. Visualization and alert generation using bounding boxes and timestamps

Each stage is modular and integrated into a structured pipeline to process incoming CCTV footage in real-time or batch.

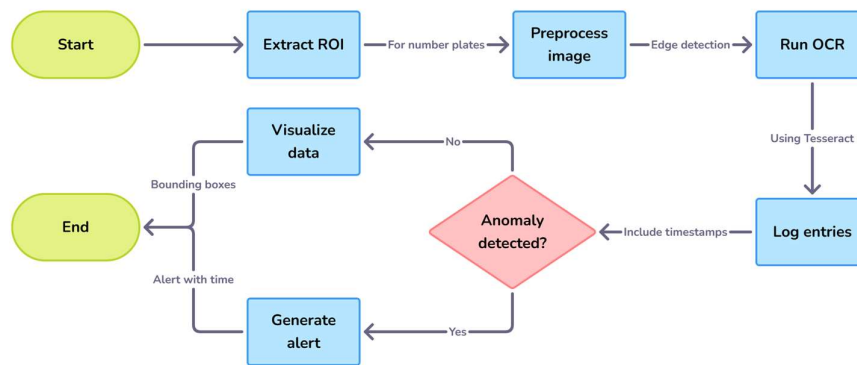


Fig 3.1 Process Flow for vehicle anomaly detection

3.3. Description of Algorithms Used

1. YOLOv8 (You Only Look Once)

YOLOv8 is a state-of-the-art object detection algorithm known for its speed and accuracy. In this project, it identifies vehicles in CCTV frames and provides bounding box coordinates for further processing.

2. Optical Character Recognition (OCR)

Tesseract and EasyOCR are employed to read alphanumeric text from license plates. EasyOCR performs better in low-light or distorted conditions. Before OCR, image preprocessing includes grayscale conversion, bilateral filtering, and Canny edge detection.

3. Homography Transformation (Optional)

If the camera view is tilted or oblique, a homography matrix is used to map the plate region to a flat, planar view, thereby improving OCR accuracy and bounding box precision.

4. Z-score Statistical Analysis

Z-score is used to detect outliers in entry time or duration of stay: High absolute Z-scores flag entries that deviate significantly from the norm, e.g., a vehicle entering at midnight repeatedly.

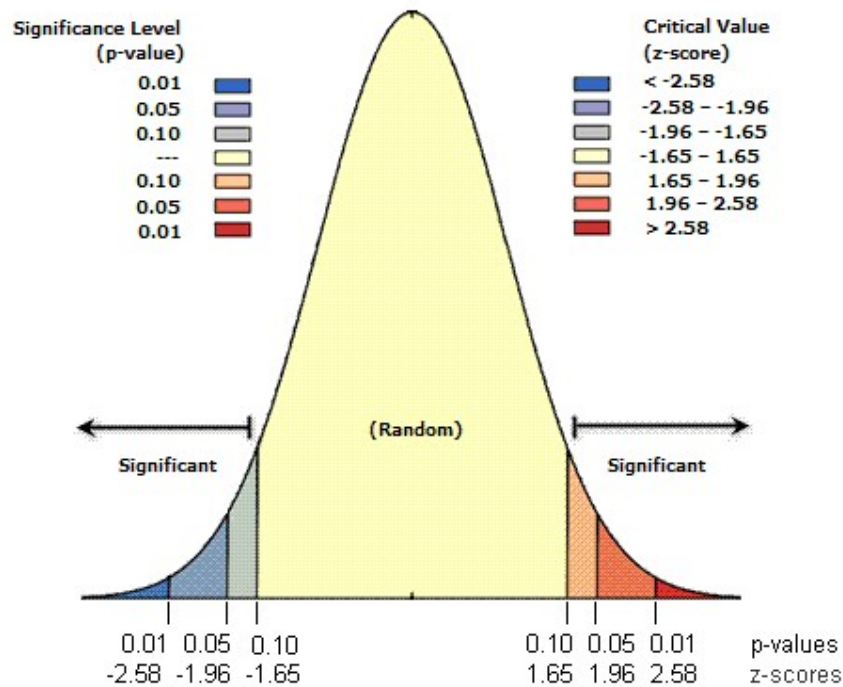


Figure 3.2 Z-Score Statistical Analysis

5. Isolation Forest

This unsupervised machine learning algorithm isolates anomalies by constructing decision trees. It detects:

- Unusual frequency of vehicle re-entry
- Long durations of stay
- Sparse or unique patterns in vehicle activity logs

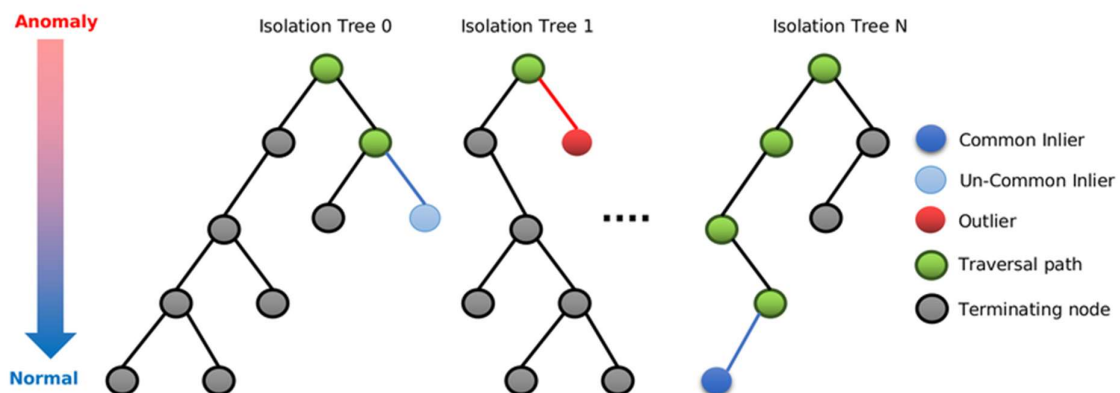


Figure 3.3 Isolation Forest Algorithm

6. Regular Expression Matching

Regex is used to validate and sanitize OCR outputs. The pattern used ensures the detected license plate is in a standard format: `^[A-Z]{2}[0-9]{2}[A-Z]{1,2}[0-9]{4}$` This prevents false positives from non-standard or unreadable plates.

4. Code Implementation

- Vehicle Detection

YOLOv8 is used to detect vehicles frame-by-frame. Once vehicles are detected, the bounding box coordinates are extracted.

- ROI Extraction and Preprocessing

From each vehicle, a subregion (ROI) containing the potential license plate is cropped. This ROI is then converted to grayscale, filtered using bilateral filtering to preserve edges while reducing noise, and finally passed through Canny edge detection.

- OCR and Validation

OCR libraries (Tesseract and EasyOCR) are used to extract alphanumeric content from the license plate region. OCR results are filtered using regex for format validation and stored in logs.

- Anomaly Detection

Z-score analysis detects statistical outliers in time of entry and duration.

Isolation Forest identifies data points that differ significantly from the majority.

- Logging and Alerting

Detected anomalies are logged in CSV files. Video timestamps around anomalies (± 5 seconds) are extracted for review. Alerts are generated with bounding boxes color-coded (red for anomaly, green for normal).

3.5. Results

- Real-time detection achieved at ~20 FPS.
- OCR accuracy was approximately 85% under standard lighting.
- Successfully detected:
 - Vehicles entering frequently within short intervals

- License plates with formatting anomalies
- Suspiciously timed entries (e.g., early morning hours)
- Alerts were visually displayed and logged for backend access.

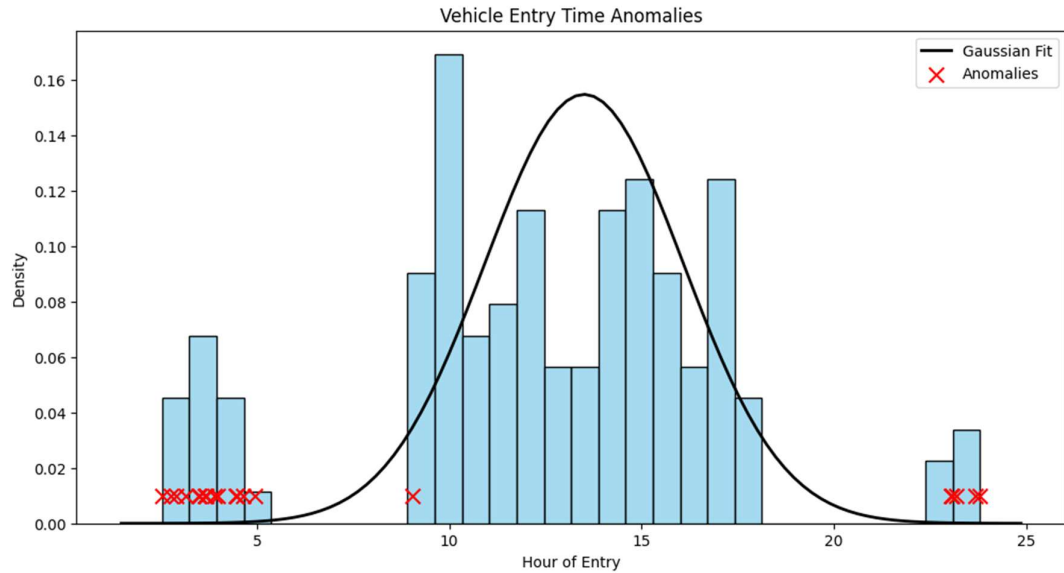


Figure 3.4 Z-score Distribution

Plate	Confidence	Direction	Z_Score	Z_Anomaly	clip_start	clip_end
MC862BE	0.96	in	-4.16408	TRUE	25-06-2025 02:46	25-06-2025 02:46
NN933GX	0.74	in	-4.11884	TRUE	25-06-2025 02:53	25-06-2025 02:53
TA510WJ	0.79	in	-3.43366	TRUE	26-06-2025 04:39	26-06-2025 04:39
FX726WT	0.79	in	3.709012	TRUE	27-06-2025 23:03	27-06-2025 23:04
ZB278CP	0.82	in	-3.88613	TRUE	27-06-2025 03:28	27-06-2025 03:29

Table 3.1 Z-Anomaly Detection

3.6. Problems Faced

- OCR reliability dropped under poor lighting or with obscured plates.
- Camera blur and frame skips caused partial detections.
- Vehicles with non-standard number plates were frequently misclassified.
- Delay in timestamp syncing caused issues with video clip generation.

3.7. Future Scope

- Integrate deep learning for end-to-end number plate detection and classification.
- Use LSTM networks for temporal behavior modeling.
- Expand the system to support multiple synchronized CCTV feeds.
- Implement cloud-based dashboards for real-time remote monitoring.
- Include driver face recognition and automatic gate control.

CONCLUSION

The integrated project work on “Intelligent Forensic Survey of Vehicular Motion and Impact Zones using Aerial Imagery” encompasses three core modules — Vehicle Speed Detection, Structural Damage Assessment, and Vehicle Anomaly Detection through CCTV Footage — leveraging advanced computer vision, machine learning, and image processing algorithms.

Through the Vehicle Speed Detection module, we successfully demonstrated a system that can detect and estimate the speed of vehicles from aerial drone footage using YOLO for vehicle localization, homography for real-world scale transformation, and Kalman filtering for smoothing speed estimates. This solution provides a non-invasive, scalable approach for traffic surveillance and enforcement.

The Damage Assessment module used satellite and aerial imagery in conjunction with algorithms like SIFT, image differencing, and Canny-Hough transformations to detect changes in infrastructure post-disaster or post-conflict. By comparing pre- and post-event imagery, the system can flag regions of interest for rapid disaster response and structural integrity monitoring.

The Vehicle Anomaly Detection system, implemented using CCTV feeds, focuses on detecting unusual vehicular behavior patterns. With techniques like license plate recognition (OCR), Z-score analysis, Isolation Forest, and regex validation, it identifies anomalies such as frequent re-entry, loitering, or tampered plates. The pipeline ensures efficient real-time monitoring and logging, making it practical for smart city surveillance networks.

Together, these systems demonstrate the potential of aerial and stationary visual data combined with AI-driven analysis to enhance situational awareness, forensic auditing, and proactive security measures. These tools can aid traffic management authorities, disaster response units, and defense research units like DRDO in creating data-backed, intelligent monitoring systems.

Future enhancements may include integration of drone flight autonomy, cloud dashboards, transformer-based tracking models, and multimodal analysis to broaden scope and accuracy. This project lays the groundwork for a robust, real-time, scalable surveillance and assessment platform.

REFERENCES

1. Bochkovskiy, A., Wang, C. Y., & Liao, H. Y. M. (2020). YOLOv4: Optimal Speed and Accuracy of Object Detection. arXiv:2004.10934.
2. Jocher, G., et al. (2023). Ultralytics YOLOv8 GitHub Repository. <https://github.com/ultralytics/ultralytics>
3. Bradski, G. (2000). The OpenCV Library. Dr. Dobb's Journal of Software Tools.
4. OpenCV Documentation. <https://docs.opencv.org>
5. JaidevAI. EasyOCR GitHub Repository. <https://github.com/JaidevAI/EasyOCR>
6. Lowe, D. G. (2004). Distinctive Image Features from Scale-Invariant Keypoints (SIFT). International Journal of Computer Vision.
7. Canny, J. (1986). A Computational Approach to Edge Detection. IEEE Transactions on Pattern Analysis and Machine Intelligence.
8. Duda, R. O., & Hart, P. E. (1972). Use of the Hough Transformation to Detect Lines and Curves in Pictures. Communications of the ACM.